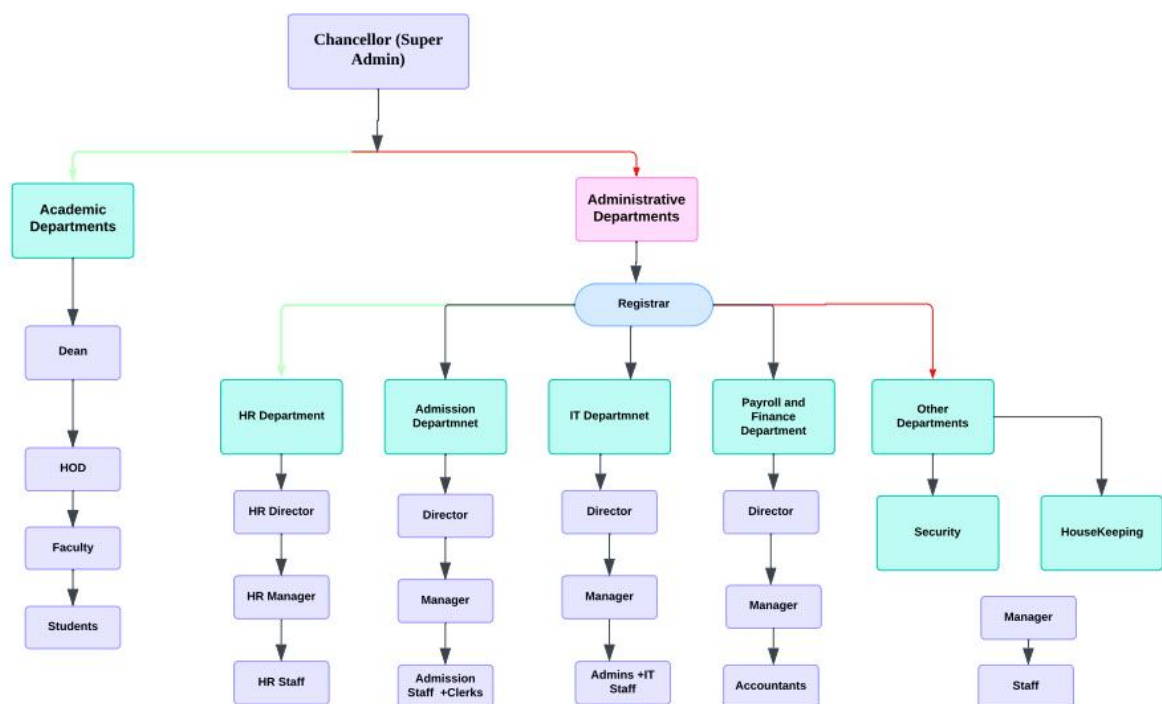


University Management System Proposal

Overview

The University Management System (UMS) is a sophisticated platform designed to streamline academic and administrative workflows within the university. The system integrates various departments, ensuring seamless communication, efficient processes, and centralized data management. Key functionalities include managing student lifecycle, faculty operations, payroll, shifts, and attendance, along with enhanced reporting and analytics.

Hierarchy:



Module 1: User Management Module

Purpose: Centralized handling of user roles, permissions, authentication, and lifecycle management for all system users, including **students**, **faculty**, and **staff**.

Key Functionalities

1.1. User Registration

- Enables creation of user accounts (students, faculty, staff) with role allocation and basic details.
- **CRUD (Create, Read, Update, Delete):**
 - **Super Admin:** Handles all user registrations.
 - **Registrar:** Manages registration for **students** and **faculty**.
 - **HR Department:** Handles registration for **employees**.

1.2. User Profile Management

- Allows authorized users to create, view, update, and delete user profiles.
- **CRUD:**
 - **Super Admin:** Full access to all profiles.
 - **Registrar:** Updates student and faculty profiles.
 - **HR Department:** Updates staff/employee profiles.
 - **Department Directors:** Approve and manage faculty updates.

1.3. Role-Based Access Control (RBAC)

- Grants permissions based on roles (e.g., Super Admin, Registrar, Faculty, Student).
- **CRUD:**
 - **Super Admin:** Assigns and updates roles for all users.
 - **IT Department:** Implements and audits access control.

1.4. User Authentication and Authorization

- Ensures secure login and system access based on assigned roles.
- **CRUD:**
 - **IT Department:** Manages system authentication and authorization.

1.5. User Password Management

- Allows secure password creation, resets, and updates for all users.
- **CRUD:**
 - **IT Department:** Handles password resets and security enforcement.

Roles & Responsibilities

- **Super Admin**

- Full oversight of user accounts, roles, permissions, and CRUD operations.
- **Reports to See:**
 - **User Registration Report**
 - **User Profile Update Report**
 - **Role-Based Access Report**
 - **User Authentication Log**
 - **Password Reset Activity Report**
 - **User Deletion Report**

- **Registrar**

- Manages **student** and **faculty** registrations, profile updates, and lifecycle data.
- **Reports to See:**
 - **User Registration Report**
 - **User Profile Update Report**
 - **User Deletion Report**
 - **Role-Based Access Report**

- **HR Department**

- Handles **employee** (administrative staff) profiles, updates, and role assignments.
- **Reports to See:**
 - **User Registration Report**
 - **User Profile Update Report**
 - **User Deletion Report**
 - **Role-Based Access Report**

- **Department Directors**

- Approve and validate role assignments for faculty under their departments.
- **Reports to See:**
 - **User Registration Report**

- **User Profile Update Report**
- **User Deletion Report**
- **IT Department**
 - Ensures secure system authentication, access control, and data protection.
 - **Reports to See:**
 - **User Authentication Log**
 - **Password Reset Activity Report**
 - **Role-Based Access Report**
 - **User Deletion Report**
- **Faculty**
 - Limited access to manage their own profiles and view student data under their supervision.
 - **Reports to See:**
 - **User Profile Update Report**
 - **User Deletion Report**
- **Students**
 - Limited access to view and update their personal profiles.
 - **Reports to See:**
 - **User Profile Update Report**
 - **User Deletion Report**

Required Reports

- **User Registration Report**
 - Tracks newly registered students, faculty, and staff with roles and departments.
- **User Profile Update Report**
 - Logs profile updates made for students, faculty, and staff.
- **Role-Based Access Report**
 - Summarizes access permissions granted to each role.

- **User Authentication Log**

- Records login attempts and access history for all users.

- **Password Reset Activity Report**

- Details password changes, resets, and failed recovery attempts.

- **User Deletion Report**

- Lists users removed from the system with reasons for deletion.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
Students	Create, Read, Update, Delete	Registrar, Super Admin
Faculty	Create, Read, Update, Delete	Registrar, Department Directors
Employees	Create, Read, Update, Delete	HR Department, Super Admin
System Users (All)	Authentication, Access	IT Department, Super Admin

Responsible Departments

- **Registrar:** Handles student and faculty registration and updates.
- **HR Department:** Manages employee user lifecycle and data.
- **IT Department:** Ensures secure system authentication and access.
- **Department Directors:** Validate and approve faculty data updates.

Module 2: Employee Management Module

Purpose: Efficient management of employee data, including their profiles, job assignments, attendance, payroll, and performance. This module helps track employee life cycle, including registrations, leave management, shifts, and payroll processing.

Key Functionalities

2.1. Employee Registration

- **Description:** Enables the creation of employee profiles, including personal details, job title, department, and role assignment.
- **CRUD Operations:**
 - **HR Department:** Responsible for creating and managing employee profiles.
 - **Super Admin:** Provides oversight for employee registration across the system.

2.2. Employee Profile Management

- **Description:** Allows management and updates of employee information, such as job title, department, personal details, and work status.
- **CRUD Operations:**
 - **HR Department:** Updates employee profiles (e.g., promotions, transfers).
 - **Super Admin:** Has full access to all employee profiles for monitoring and validation.

2.3. Employee Job Assignment

- **Description:** Assigns employees to specific departments, roles, and responsibilities within the organization.
- **CRUD Operations:**
 - **HR Department:** Assigns or reassigns employees to specific roles, departments, and shifts.
 - **Department Directors:** Can approve or modify assignments within their departments.

2.4. Employee Performance Management

- **Description:** Tracks and evaluates employee performance, including goal setting, feedback, and performance reviews.
- **CRUD Operations:**
 - **HR Department:** Implements and monitors performance management processes.

- **Department Directors:** Conduct performance reviews for employees within their departments.
- **Super Admin:** Oversees overall employee performance evaluations and reports.

2.5. Employee Leave Management

- **Description:** Manages leave requests, approvals, and leave balances for employees.
- **CRUD Operations:**
 - **HR Department:** Manages and approves leave requests for employees.
 - **Department Directors:** Approve or reject leave requests for employees under their management.

Roles & Responsibilities

• Super Admin

- **Responsibilities:** Oversees all aspects of employee management, including registration, profile updates, job assignments, performance, and leave management.
- **Reports to See:**
 - **Employee Registration Report**
 - **Employee Profile Update Report**
 - **Employee Leave Report**
 - **Employee Performance Report**

• HR Department

- **Responsibilities:** Handles employee registrations, updates, job assignments, performance tracking, and leave management.
- **Reports to See:**
 - **Employee Registration Report**
 - **Employee Profile Update Report**
 - **Employee Leave Report**
 - **Employee Performance Report**

• Department Directors

- **Responsibilities:** Manage employee performance and attendance, assign jobs, and approve leaves within their departments.
- **Reports to See:**
 - **Employee Job Assignment Report**
 - **Employee Performance Report**
 - **Employee Leave Report**
- **Employees**
 - **Responsibilities:** Manage their own profiles, submit leave requests, and participate in performance evaluations.
 - **Reports to See:**
 - **Employee Profile Update Report**
 - **Employee Leave Report**
 - **Employee Performance Report**

Required Reports

- **Employee Registration Report**
 - Tracks newly registered employees with details such as role, department, and status.
- **Employee Profile Update Report**
 - Logs any updates or changes made to employee profiles.
- **Employee Leave Report**
 - Provides a summary of leave requests, approvals, and leave balances for employees.
- **Employee Performance Report**
 - Tracks employee performance, goals, feedback, and evaluations.
- **Employee Job Assignment Report**
 - Provides details about job assignments, role changes, and department allocations for employees.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
Employees	Create, Read, Update, Delete	HR Department, Super Admin
Department Directors	Approve, Update	HR Department, Super Admin

Responsible Departments

- **HR Department:** Handles all employee registrations, updates, job assignments, performance tracking, and leave management.
- **Department Directors:** Manage performance, attendance, and job assignments within their departments.
- **Super Admin:** Oversees all employee management activities across the system.

Module 3: Attendance Management Module

Purpose: Streamlines the tracking and management of attendance for students, faculty, and employees, ensuring accurate records for performance evaluation, payroll integration, and analytics.

Key Functionalities

3.1. Attendance Tracking

- **Description:** Tracks attendance using automated systems (e.g., biometric integration) or manual entry for exceptional cases.
- **CRUD Operations:**
 - **HR Department:** Manages employee attendance tracking and corrections.
 - **Registrar:** Oversees manual attendance tracking for students and faculty.
 - **Department Directors:** Monitors attendance trends for their departments.

3.2. Attendance Reporting

- **Description:** Generates reports summarizing attendance records, such as total presence, absence, and trends for all user categories.
- **CRUD Operations:**
 - **HR Department:** Prepares attendance reports for employees.
 - **Registrar:** Prepares attendance reports for students and faculty.

3.3. Attendance Analytics

- **Description:** Provides insights into attendance trends, including absenteeism rates and compliance with schedules.
- **CRUD Operations:**
 - **HR Department:** Analyzes employee attendance data for payroll and performance.
 - **Registrar:** Analyzes attendance trends for students and faculty.

3.4. Attendance Approval Workflow

- **Description:** Allows managers to approve or reject attendance corrections or exceptions submitted by employees or faculty.
- **CRUD Operations:**
 - **HR Department:** Approves corrections for employees.
 - **Department Directors:** Approve corrections for faculty under their departments.
 - **Registrar:** Approves corrections for student attendance.

3.5. Attendance Integration with Payroll

- **Description:** Links attendance data with payroll processing to calculate salaries based on hours worked or absenteeism.
- **CRUD Operations:**
 - **HR Department:** Validates attendance data for payroll calculations.
 - **IT Department:** Maintains integration between attendance and payroll systems.

Roles & Responsibilities

- **Super Admin**

- **Responsibilities:** Monitors overall attendance trends across all departments.
- **Reports to See:**
 - **Attendance Overview Report**
 - **Attendance Exception Report**
 - **Attendance-Payroll Integration Report**

- **HR Department**

- **Responsibilities:** Tracks, approves, and reports attendance for employees; validates attendance data for payroll.
- **Reports to See:**
 - **Employee Attendance Report**
 - **Attendance Approval Report**
 - **Attendance-Payroll Report**

- **Registrar**

- **Responsibilities:** Oversees student and faculty attendance, manages manual corrections, and generates reports.
- **Reports to See:**
 - **Student Attendance Report**
 - **Faculty Attendance Report**
 - **Attendance Trends Report**

- **Department Directors**

- **Responsibilities:** Monitor and approve attendance for faculty and department staff.
- **Reports to See:**
 - **Faculty Attendance Report**
 - **Department Attendance Trends Report**

- **Employees and Faculty**

- **Responsibilities:** Submit corrections or exceptions for attendance records; view attendance history.
- **Reports to See:**
 - **Individual Attendance Report**

- **Students**

- **Responsibilities:** View their own attendance records and ensure compliance with attendance policies.
- **Reports to See:**
 - **Individual Attendance Report**

Required Reports

- **Attendance Overview Report**

- Summarizes attendance rates across all user groups (students, faculty, employees).

- **Attendance Exception Report**

- Tracks cases of manual corrections or exceptions made to attendance records.

- **Attendance Trends Report**

- Analyzes trends in attendance over time for departments, employees, and students.

- **Employee Attendance Report**

- Provides detailed attendance records for all employees, including shift-wise breakdowns.

- **Faculty Attendance Report**

- Offers insights into faculty attendance patterns and compliance.

- **Student Attendance Report**

- Summarizes student attendance by course, class, and department.

- **Attendance Approval Report**

- Tracks the status of approved or rejected attendance corrections.

- **Attendance-Payroll Integration Report**

- Links attendance data to payroll calculations for employees.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
Employees	Read, Update (Corrections)	HR Department, Super Admin
Faculty	Read, Update (Corrections)	Registrar, Department Directors
Students	Read	Registrar, Super Admin

Responsible Departments

- **HR Department:** Manages attendance tracking and approval for employees.
- **Registrar:** Oversees student and faculty attendance tracking and reporting.
- **Department Directors:** Monitor attendance trends and handle approvals for their departments.
- **IT Department:** Ensures seamless integration of attendance with payroll systems.

Module 4: Payroll Management Module

Purpose: Facilitates accurate and efficient salary processing by integrating attendance, leave, and performance data, ensuring compliance with organizational and statutory regulations.

Key Functionalities

4.1. Payroll Processing

- **Description:** Automates salary calculations based on attendance, shifts, leaves, and deductions.
- **CRUD Operations:**
 - **HR Department:** Responsible for generating and verifying payroll.
 - **Finance Department:** Validates and finalizes salary disbursements.

4.2. Payroll Reporting

- **Description:** Provides detailed reports on payroll, including salary components, deductions, and tax calculations.
- **CRUD Operations:**
 - **HR Department:** Generates initial payroll reports.
 - **Finance Department:** Reviews reports for accuracy before salary disbursement.

4.3. Payroll Analytics

- **Description:** Analyzes payroll data to provide insights into labor costs, salary trends, and compliance adherence.
- **CRUD Operations:**
 - **HR Department:** Tracks analytics for decision-making.
 - **Finance Department:** Uses analytics for budgeting and financial forecasting.

4.4. Payroll Compliance Management

- **Description:** Ensures payroll adheres to tax regulations, labor laws, and organizational policies.
- **CRUD Operations:**
 - **HR Department:** Manages compliance with employee policies.
 - **Finance Department:** Handles tax filings and compliance with statutory laws.

4.5. Payroll Integration with Attendance

- **Description:** Links attendance records with payroll processing to ensure accurate salary calculations based on hours worked and leaves.
- **CRUD Operations:**
 - **IT Department:** Maintains integration systems.
 - **HR Department:** Monitors data consistency.

Roles & Responsibilities

- **Super Admin**
 - **Responsibilities:** Oversees payroll processes and monitors compliance across departments.

- **Reports to See:**
 - **Comprehensive Payroll Summary Report**
 - **Payroll Compliance Report**
- **HR Department**
 - **Responsibilities:** Manages payroll processing, reporting, compliance, and integration with attendance data.
 - **Reports to See:**
 - **Employee Payroll Report**
 - **Payroll Attendance Report**
 - **Payroll Analytics Report**
- **Finance Department**
 - **Responsibilities:** Reviews payroll reports, handles compliance filings, and ensures timely salary disbursements.
 - **Reports to See:**
 - **Salary Disbursement Report**
 - **Payroll Tax Compliance Report**
 - **Payroll Forecasting Report**
- **Employees and Faculty**
 - **Responsibilities:** View salary breakdowns and report discrepancies.
 - **Reports to See:**
 - **Individual Payroll Slip**

Required Reports

- **Comprehensive Payroll Summary Report**
 - Summarizes payroll data, including total salaries, deductions, and disbursed amounts.
- **Employee Payroll Report**
 - Details individual employee salaries, deductions, bonuses, and net payments.

- **Payroll Attendance Report**

- Links attendance and payroll to provide insights into salary calculations based on presence and absence.

- **Payroll Analytics Report**

- Tracks labor costs, trends, and deviations over time.

- **Payroll Compliance Report**

- Verifies adherence to tax laws, statutory deductions, and organizational policies.

- **Salary Disbursement Report**

- Summarizes details of salary payments, including bank transfers and pending disbursements.

- **Payroll Tax Compliance Report**

- Provides insights into tax filings and compliance-related deductions.

- **Payroll Forecasting Report**

- Analyzes payroll data to predict future labor costs and budget allocations.

- **Individual Payroll Slip**

- Detailed salary breakdown for each employee or faculty member.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
Employees/Faculty	Read	HR Department, Finance Department
HR Department	Create, Read, Update	Super Admin, Finance Department
Finance Department	Approve, Read	Super Admin

Responsible Departments

- **HR Department:** Handles payroll processing, reporting, analytics, and compliance.
- **Finance Department:** Validates reports, manages tax compliance, and processes salary disbursements.
- **IT Department:** Maintains system integrations between payroll and attendance modules.
- **Super Admin:** Oversees the entire payroll management system.

Module 5: Shift Management Module

Purpose: Ensures effective allocation, scheduling, and management of employee shifts to optimize organizational productivity and meet operational demands.

Key Functionalities

5.1. Shift Creation and Assignment

- **Description:** Defines shifts (e.g., morning, evening, night) and assigns them to employees or departments.
- **CRUD Operations:**
 - **HR Department:** Creates and assigns shifts for employees.
 - **Department Directors:** Assign shifts to departmental staff.

5.2. Shift Scheduling

- **Description:** Manages detailed schedules for employees, ensuring proper staffing and compliance with organizational policies.
- **CRUD Operations:**
 - **HR Department:** Sets and adjusts master shift schedules.
 - **Department Directors:** Adjust schedules as per departmental requirements.

5.3. Shift Reporting

- **Description:** Generates detailed reports on shift allocations, employee adherence, and any gaps or conflicts in coverage.
- **CRUD Operations:**
 - **HR Department:** Prepares reports for organization-wide shifts.
 - **Department Directors:** Generate department-specific shift reports.

5.4. Shift Analytics

- **Description:** Analyzes shift data to optimize staffing, identify overburdened employees, and maintain productivity levels.
- **CRUD Operations:**
 - **HR Department:** Monitors and analyzes organization-wide shift data.
 - **Department Directors:** Use analytics for department-specific staffing adjustments.

5.5. Shift Integration with Attendance and Payroll

- **Description:** Links shifts with attendance and payroll modules to calculate work hours, overtime, and compliance.
- **CRUD Operations:**
 - **IT Department:** Maintains system integration.
 - **HR Department:** Ensures accurate data flow between modules.

Roles & Responsibilities

• Super Admin

- **Responsibilities:** Monitors overall shift schedules and ensures compliance with organizational policies.
- **Reports to See:**
 - **Shift Coverage Summary Report**
 - **Shift Utilization Report**

• HR Department

- **Responsibilities:** Creates and manages master shift schedules, assigns shifts, and analyzes organizational requirements.
- **Reports to See:**
 - **Organization-Wide Shift Report**

- **Shift Conflicts Report**
- **Department Directors**
 - **Responsibilities:** Assign shifts to departmental employees, track adherence, and resolve scheduling conflicts.
 - **Reports to See:**
 - **Department Shift Allocation Report**
 - **Shift Compliance Report**
- **Employees and Faculty**
 - **Responsibilities:** View their assigned shifts and report any conflicts or issues.
 - **Reports to See:**
 - **Individual Shift Assignment Report**

Required Reports

- **Shift Coverage Summary Report**
 - Provides an overview of coverage across shifts, identifying under- or over-staffed periods.
- **Organization-Wide Shift Report**
 - Details all assigned shifts, including employee assignments and schedules.
- **Shift Conflicts Report**
 - Highlights overlapping or unassigned shifts to prevent staffing gaps.
- **Shift Utilization Report**
 - Analyzes efficiency of staffing levels and adherence to scheduled shifts.
- **Department Shift Allocation Report**
 - Summarizes shift assignments within a specific department.

- **Shift Compliance Report**

- Tracks employee compliance with assigned shifts and identifies any discrepancies.

- **Individual Shift Assignment Report**

- Details the shifts assigned to an individual employee or faculty member.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
Employees/Faculty	Read, Update (Conflicts)	HR Department, Department Directors
HR Department	Create, Read, Update	Super Admin, IT Department
Department Directors	Create, Read, Update	HR Department

Responsible Departments

- **HR Department:** Oversees shift creation, assignment, and reporting for employees.
- **Department Directors:** Handle shift management for their departmental staff.
- **IT Department:** Maintains integration between shifts, attendance, and payroll modules.
- **Super Admin:** Monitors compliance and resolves high-level conflicts.

Module 6: Leave Management Module

Purpose: Simplifies the process of managing leave applications, approvals, and tracking for employees and faculty.

Key Functionalities

6.1. Leave Policy Management

- **Description:** Defines rules for leave types (e.g., annual, sick, casual) and policies for approval.
- **CRUD Operations:**
 - **HR Department:** Creates and updates leave policies.
 - **Super Admin:** Reviews and approves organizational policies.

6.2. Leave Application and Approval

- **Description:** Allows employees and faculty to apply for leave and managers to approve or reject requests.
- **CRUD Operations:**
 - **Employees/Faculty:** Submit leave applications.
 - **Department Directors/Managers:** Approve or reject leave requests.

6.3. Leave Tracking and Reporting

- **Description:** Tracks leave balances and generates reports on leave utilization for departments or individuals.
- **CRUD Operations:**
 - **HR Department:** Monitors leave usage and generates reports.
 - **Super Admin:** Reviews overall leave trends.

6.4. Leave Analytics

- **Description:** Analyzes leave trends to identify patterns like frequent absences or overused leave.
- **CRUD Operations:**
 - **HR Department:** Tracks leave data for decision-making.
 - **Super Admin:** Uses analytics for organizational insights.

6.5. Leave Integration with Payroll

- **Description:** Links leave records with payroll to calculate deductions or payouts for unused leave.
- **CRUD Operations:**
 - **IT Department:** Ensures smooth integration of leave data with payroll systems.

- **HR Department:** Ensures accurate records are reflected in payroll.

Roles & Responsibilities

- **Super Admin**

- **Responsibilities:** Oversees leave policies and monitors trends.
- **Reports to See:**
 - **Leave Utilization Summary**
 - **Organization-Wide Leave Trends Report**

- **HR Department**

- **Responsibilities:** Sets leave policies, tracks leave balances, and generates detailed reports.
- **Reports to See:**
 - **Departmental Leave Usage Report**
 - **Leave Policy Compliance Report**

- **Department Directors/Managers**

- **Responsibilities:** Approve or reject leave applications for their teams and manage coverage during absences.
- **Reports to See:**
 - **Individual Leave Usage Report**
 - **Leave Application Status Report**

- **Employees/Faculty**

- **Responsibilities:** Apply for leave and view their leave balance.
- **Reports to See:**
 - **Personal Leave Balance Report**

Required Reports

- **Leave Utilization Summary**

- Tracks overall leave usage across the organization.

- **Organization-Wide Leave Trends Report**

- Analyzes leave patterns, highlighting peak periods or frequent absences.

- **Departmental Leave Usage Report**

- Summarizes leave balances and usage for employees within a department.

- **Leave Policy Compliance Report**

- Tracks adherence to leave policies, such as maximum allowable leaves.

- **Individual Leave Usage Report**

- Details leave taken by a specific employee or faculty member.

- **Leave Application Status Report**

- Shows the status of submitted leave requests (approved, pending, or rejected).

- **Personal Leave Balance Report**

- Displays remaining leave balance for individual employees or faculty.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
Employees/Faculty	Create (Apply), Read	HR Department, Managers
HR Department	Create, Read, Update	Super Admin
Department Directors/Managers	Read, Update	HR Department

Responsible Departments

- **HR Department:** Manages policies, tracking, and reporting of leave.

- **Department Directors/Managers:** Approve or reject leave requests and ensure proper coverage.
- **Super Admin:** Monitors leave trends and policy compliance.
- **IT Department:** Ensures integration between leave and payroll systems.

Module 7: Performance Management Module

Purpose: Helps track, evaluate, and improve employee and faculty performance to support organizational growth and personal development.

Key Functionalities

7.1. Performance Goal Setting

- **Description:** Allows managers to define individual and team performance goals.
- **CRUD Operations:**
 - **Managers:** Create and assign performance goals.
 - **HR Department:** Monitors overall goal-setting compliance.

7.2. Performance Evaluation and Feedback

- **Description:** Conducts appraisals based on set goals and provides structured feedback.
- **CRUD Operations:**
 - **Managers:** Evaluate and record feedback for their teams.
 - **HR Department:** Reviews and validates evaluation processes.

7.3. Performance Reporting and Analytics

- **Description:** Tracks performance metrics and generates reports for analysis.
- **CRUD Operations:**
 - **HR Department:** Prepares reports for decision-making.
 - **Super Admin:** Reviews organization-wide performance trends.

7.4. Performance Integration with Payroll and Promotions

- **Description:** Links performance outcomes with salary increments, bonuses, and promotions.
- **CRUD Operations:**
 - **HR Department:** Ensures accurate integration with payroll.

- **IT Department:** Maintains system connections.

Roles & Responsibilities

- **Super Admin**

- **Responsibilities:** Monitors overall performance management and ensures alignment with organizational goals.
- **Reports to See:**
 - **Organizational Performance Overview Report**
 - **High-Performer and Low-Performer Trends**

- **HR Department**

- **Responsibilities:** Facilitates the performance management process, monitors goals, and analyzes outcomes.
- **Reports to See:**
 - **Performance Goal Tracking Report**
 - **Appraisal Process Compliance Report**

- **Department Directors/Managers**

- **Responsibilities:** Set goals, evaluate performance, and provide actionable feedback for their teams.
- **Reports to See:**
 - **Individual Performance Review Report**
 - **Departmental Performance Trends Report**

- **Employees/Faculty**

- **Responsibilities:** Review assigned goals, provide self-assessments, and act on feedback.
- **Reports to See:**
 - **Personal Performance Feedback Report**

Required Reports

- **Organizational Performance Overview Report**

- Provides a summary of performance metrics across the organization.
- **High-Performer and Low-Performer Trends**
 - Highlights patterns of top and underperformers to assist in recognition or improvement planning.
- **Performance Goal Tracking Report**
 - Monitors the progress of performance goals at individual and team levels.
- **Appraisal Process Compliance Report**
 - Tracks completion rates of evaluations and feedback cycles.
- **Individual Performance Review Report**
 - Details evaluations and feedback for a specific employee or faculty member.
- **Departmental Performance Trends Report**
 - Summarizes performance trends and patterns within specific departments.
- **Personal Performance Feedback Report**
 - Provides detailed performance reviews and feedback for an individual.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
Employees/Faculty	Read, Update (Self-Evaluations)	Managers, HR Department
HR Department	Create, Read, Update	Super Admin
Managers	Create, Read, Update	HR Department

Responsible Departments

- **HR Department:** Oversees goal setting, evaluations, and integration with payroll/promotions.
- **Department Directors/Managers:** Conduct evaluations and provide feedback for their teams.
- **Super Admin:** Reviews and monitors performance processes at a high level.
- **IT Department:** Manages system integration with payroll and other relevant modules.

Module 8: Reporting and Analytics Module

Purpose: Provides clear and detailed reports and insights to help in decision-making and tracking organizational performance.

Key Functionalities

8.1. Customizable Reporting

- **Description:** Allows users to create custom reports based on their specific needs.
- **CRUD Operations:**
 - **HR Department:** Creates and customizes reports for employees and departments.
 - **Super Admin:** Monitors and approves critical reports.

8.2. Real-Time Analytics

- **Description:** Shows real-time data and trends, such as attendance, performance, or payroll statistics.
- **CRUD Operations:**
 - **HR Department:** Monitors live analytics for employees and organizational data.
 - **Super Admin:** Uses analytics to review key performance metrics.

8.3. Data Visualization

- **Description:** Presents data in easy-to-understand formats like charts, graphs, and tables.
- **CRUD Operations:**

- **IT Department:** Manages and updates visualization tools.
- **HR and Managers:** View and use visualized data for decision-making.

8.4. Reporting and Analytics Integration with Other Modules

- **Description:** Combines data from different modules, such as attendance, payroll, or performance, into a single report.
- **CRUD Operations:**
 - **IT Department:** Maintains integration across modules.
 - **HR Department:** Generates consolidated reports.

Roles & Responsibilities

• Super Admin

- **Responsibilities:** Monitors overall reporting and ensures data accuracy.
- **Reports to See:**
 - **Organizational Insights Report**
 - **Module Integration Overview**

• HR Department

- **Responsibilities:** Prepares detailed reports for departments and ensures correct data inputs.
- **Reports to See:**
 - **Employee Performance Report**
 - **Attendance and Payroll Summary**

• Department Directors/Managers

- **Responsibilities:** Review departmental reports and track team performance.
- **Reports to See:**
 - **Departmental Analytics Report**
 - **Team Performance Trends Report**

• IT Department

- **Responsibilities:** Manages system reliability and ensures smooth data integration.
- **Reports to See:**

- **System Health and Data Flow Report**

Required Reports

- **Organizational Insights Report**

- Summarizes key organizational metrics, like attendance, payroll, and performance.

- **Module Integration Overview**

- Shows how data flows between modules and highlights any gaps or errors.

- **Employee Performance Report**

- Tracks individual and team performance across the organization.

- **Attendance and Payroll Summary**

- Combines attendance and payroll data to show how work hours affect payouts.

- **Departmental Analytics Report**

- Provides insights specific to individual departments, such as staffing trends.

- **Team Performance Trends Report**

- Highlights team-level performance metrics over time.

- **System Health and Data Flow Report**

- Details the status of system integrations and identifies any technical issues.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
------------	-----------------	------------

HR Department	Create, Read, Update	Super Admin, IT Department
Managers	Read	HR Department, IT Department
Super Admin	Create, Read	IT Department
IT Department	Update, Maintain Systems	Super Admin

Responsible Departments

- **HR Department:** Creates detailed employee and departmental reports.
- **Super Admin:** Reviews high-level organizational insights and analytics.
- **Department Directors/Managers:** Use reports to track performance and departmental activities.
- **IT Department:** Ensures data accuracy and integration across modules.

Module 9: Security and Access Control Module

Purpose: Ensures the system's security by managing user access, data protection, and audit tracking.

Key Functionalities

9.1. Role-Based Access Control (RBAC)

- **Description:** Assigns permissions based on user roles, ensuring users can only access authorized features.
- **CRUD Operations:**
 - **Super Admin:** Creates, updates, and assigns roles.
 - **IT Department:** Ensures the proper implementation of roles and access.

9.2. User Authentication and Authorization

- **Description:** Verifies user identity and controls access to system modules based on their roles.
- **CRUD Operations:**
 - **Super Admin:** Configures and monitors authentication protocols.
 - **IT Department:** Manages technical implementation.

9.3. Data Encryption

- **Description:** Protects sensitive information, like payroll or performance data, by encrypting it.
- **CRUD Operations:**
 - **IT Department:** Implements and updates encryption standards.
 - **Super Admin:** Monitors encryption policies.

9.4. Access Control and Audit Trails

- **Description:** Tracks user activity and ensures compliance with data usage policies.
- **CRUD Operations:**
 - **IT Department:** Manages logs and system tracking tools.
 - **Super Admin:** Reviews audit reports for security checks.

Roles & Responsibilities

- **Super Admin**
 - **Responsibilities:** Oversees overall security measures and access protocols.
 - **Reports to See:**
 - **System Access Log Report**
 - **Role Permissions Audit Report**
- **IT Department**
 - **Responsibilities:** Implements and maintains system security features.
 - **Reports to See:**
 - **Data Encryption Status Report**
 - **Access Control Audit Report**
- **HR Department**
 - **Responsibilities:** Monitors role assignments for employees and ensures compliance with policies.
 - **Reports to See:**
 - **Role and Permission Review Report**

Required Reports

- **System Access Log Report**
 - Details all user login attempts and access activities.
- **Role Permissions Audit Report**
 - Reviews roles and their assigned permissions for potential issues.
- **Data Encryption Status Report**
 - Tracks encryption status of sensitive organizational data.
- **Access Control Audit Report**
 - Summarizes system access, highlighting any unauthorized activity.
- **Role and Permission Review Report**
 - Lists all assigned roles and verifies compliance with policies.

CRUD Summary (Create, Read, Update, Delete)

User Group	CRUD Operations	Managed By
Super Admin	Create, Read, Update	IT Department
IT Department	Create, Read, Update	Super Admin
HR Department	Read, Update	Super Admin, IT Department

Responsible Departments

- **IT Department:** Manages system security, data protection, and auditing tools.
- **Super Admin:** Ensures organizational compliance with access and security protocols.
- **HR Department:** Oversees role-based access assignments and user compliance.

Module 10: Academic Department Management

Purpose:

To manage the operations of academic departments, including department structure, faculty, courses, resources, and performance, with oversight by the Dean and other key stakeholders.

Key Functionalities

10.1. Department Creation and Management

- Enables creating, modifying, and deleting academic departments.
- CRUD (Create, Read, Update, Delete):
 - **Super Admin:** Creates and deletes departments.
 - **Dean:** Oversees department structure and ensures compliance with policies.
 - **Department Heads:** Update department-specific details like faculty rosters and facilities.
 - **Registrar:** Assists with updating department data for academic operations.

10.2. Faculty Assignment and Management

- Manages faculty assignment to departments, tracks workload, and monitors performance.
- CRUD:
 - **Super Admin:** Assigns and monitors faculty across departments.
 - **Dean:** Reviews faculty assignments and workload compliance.
 - **Department Heads:** Assign faculty roles and oversee faculty workloads.
 - **HR Department:** Assists with faculty role documentation and adjustments.

10.3. Course Management by Department

- Manages course scheduling, faculty assignments, and student enrollment tracking.
- CRUD:
 - **Super Admin:** Manages cross-departmental course offerings.

- **Dean:** Reviews departmental course offerings to ensure quality and consistency.
- **Department Heads:** Schedule courses, assign faculty, and handle department-specific curriculum needs.
- **Faculty:** View and update their assigned course schedules.

10.4. Departmental Resource Allocation

- Allocates resources such as classrooms, labs, and equipment for departmental use.
- CRUD:
 - **Super Admin:** Monitors and allocates resources across departments.
 - **Dean:** Reviews resource allocation to ensure equitable distribution.
 - **Department Heads:** Manage and allocate resources within their departments.
 - **Facilities Department:** Tracks and provides physical resources as required.

10.5. Academic Performance Tracking

- Tracks department, faculty, and course performance based on student outcomes, enrollment, and resource usage.
- CRUD:
 - **Super Admin:** Generates cross-departmental performance summaries.
 - **Dean:** Reviews performance reports and provides strategic direction.
 - **Department Heads:** Monitor departmental KPIs and take corrective actions.
 - **Registrar:** Assists in compiling academic data for performance analysis.

10.6. Department Collaboration Tools

- Provides communication and collaboration tools for department staff and faculty.
- CRUD:
 - **IT Department:** Implements tools for inter-departmental communication.
 - **Dean:** Facilitates collaboration between departments to promote synergy.
 - **Department Heads:** Coordinate discussions and team efforts for department operations.

10.7. Departmental Policy Management

- Documents and enforces academic policies specific to departments.
- CRUD:
 - **Dean:** Drafts, approves, and enforces academic policies.
 - **Department Heads:** Apply department-specific rules and regulations.
 - **Super Admin:** Ensures compliance with institutional policies.

10.8. Reporting and Analytics

- Generates reports on department operations, performance, and compliance.
- CRUD:
 - **Super Admin:** Views institution-wide analytics.
 - **Dean:** Reviews department-specific reports for decision-making.
 - **Department Heads:** Access reports for their respective departments.

Roles & Responsibilities

- **Super Admin:**
 - Oversees all academic departments, resource allocation, and performance monitoring.
 - Manages department creation, faculty assignments, and cross-departmental activities.
- **Dean:**
 - Oversees the functioning of all academic departments, reviews performance, and ensures quality standards.
 - Approves faculty assignments and reviews resource allocation.
- **Registrar:**
 - Assists with course scheduling, faculty assignments, and department performance tracking.
 - Works closely with the Dean and Department Heads for academic operations.
- **Department Heads:**
 - Manage their department's courses, faculty, resources, and operations.

- Track faculty performance and ensure compliance with academic policies.
- **HR Department:**
 - Assists in faculty role assignments, documentation, and performance tracking.
 - Coordinates with the Dean for academic staffing needs.
- **Faculty:**
 - Manage their assigned courses and contribute to department objectives.
 - Provide feedback on resource allocation and student performance.
- **Facilities Department:**
 - Tracks and provides physical resources, such as labs and classrooms, to departments.

Required Reports

- **Department Overview Report:** Tracks department details, faculty assignments, and active courses.
- **Faculty Assignment Report:** Lists faculty workloads, assignments, and associated courses.
- **Course Enrollment Report:** Summarizes student enrollment across departmental courses.
- **Department Resource Allocation Report:** Highlights usage and availability of departmental resources.
- **Academic Performance Report:** Tracks student outcomes and departmental KPIs.
- **Faculty Performance Report:** Monitors faculty workload, feedback, and contributions.
- **Inter-Departmental Collaboration Report:** Summarizes shared activities and collaborative efforts.
- **Policy Compliance Report:** Tracks adherence to academic policies.

CRUD Summary (Create, Read, Update, Delete)

<i>Entity</i>	<i>CRUD Operations</i>	<i>Managed By</i>
<i>Departments</i>	<i>Create, Read, Update, Delete</i>	<i>Super Admin, Dean, Department Heads</i>
<i>Faculty Assignments</i>	<i>Create, Read, Update, Delete</i>	<i>Super Admin, Dean, Department Heads, HR</i>
<i>Courses</i>	<i>Create, Read, Update, Delete</i>	<i>Super Admin, Dean, Department Heads</i>
<i>Reports</i>	<i>Generate, Read</i>	<i>Super Admin, Dean, Department Heads</i>
<i>Policies</i>	<i>Create, Read, Update, Delete</i>	<i>Dean, Super Admin</i>

Responsible Departments

- **Registrar:** Manages course scheduling and supports departmental operations.
- **HR Department:** Assists with faculty assignments and performance tracking.
- **Facilities Department:** Manages resources like classrooms and labs for departments.
- **IT Department:** Implements tools for department collaboration and reporting.
- **Dean's Office:** Oversees academic policies, department performance, and compliance